



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Information Technology
Academic Year 2025 - 2026 (Odd Semester)

Degree, Semester & Branch: III Semester B.Tech. IT

Course Code & Title: CS3352 & Foundations of Data Science

Name of the Faculty member (s): Dr.V. Anusuya, ASCP/IT

Innovative Practice Description

- **Unit / Topic:** Unit IV / **Data manipulation with Pandas**

- **Course Outcome:** CO4

- **Topic Learning Outcome:** TLO9

- **Activity Chosen:** One Minute Paper

- **Justification:**

In unit IV, Data Manipulations with pandas is an important topic needed to perform data analysis and lots of functions for manipulating data. This activity helps the students to recall the concepts of pandas in data manipulation.

- **Time Allotted for the Activity:** 3 Minutes

- **Details of the Implementation:**

- The topic Data manipulation with pandas was taught in the previous class.
- How to read csv file and different ways of viewing the information from dataset explained with an example.
- The time allotted for the topic is 3 Minutes.
- The students were asked to write about the topic and the understanding level as shown in Figure 1 & 2.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO4	3	3	2	2	2	1	2	2	3	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO9	PO10	PO12	PSO1	PSO2	PSO3
	3	3	2	2	2	1	2	2	3	2
Justification for correlation	Apply basic Knowledge and fundamentals of mathematics for data manipulations	The students will be able to identify the need for analysis of complex engineering problems	Able to design the process flow for developing data science applications.	The students will be able to analyze data and represent data to provide valid conclusions	The students will be able to functional individually or as a team in analyzing, representing data	The students will be able to interpret descriptions of data and prepare as a report or presentation	The students able to recognize the need for pandas for data science process.	The students able to exhibit the knowledge in data manipulation	The students can Exhibit knowledge of data science in the field of IoT, Cloud Computing to develop IT solutions	The students able to exhibit knowledge in data manipulation to develop a quality product.

• Images / Screenshot of the practice:

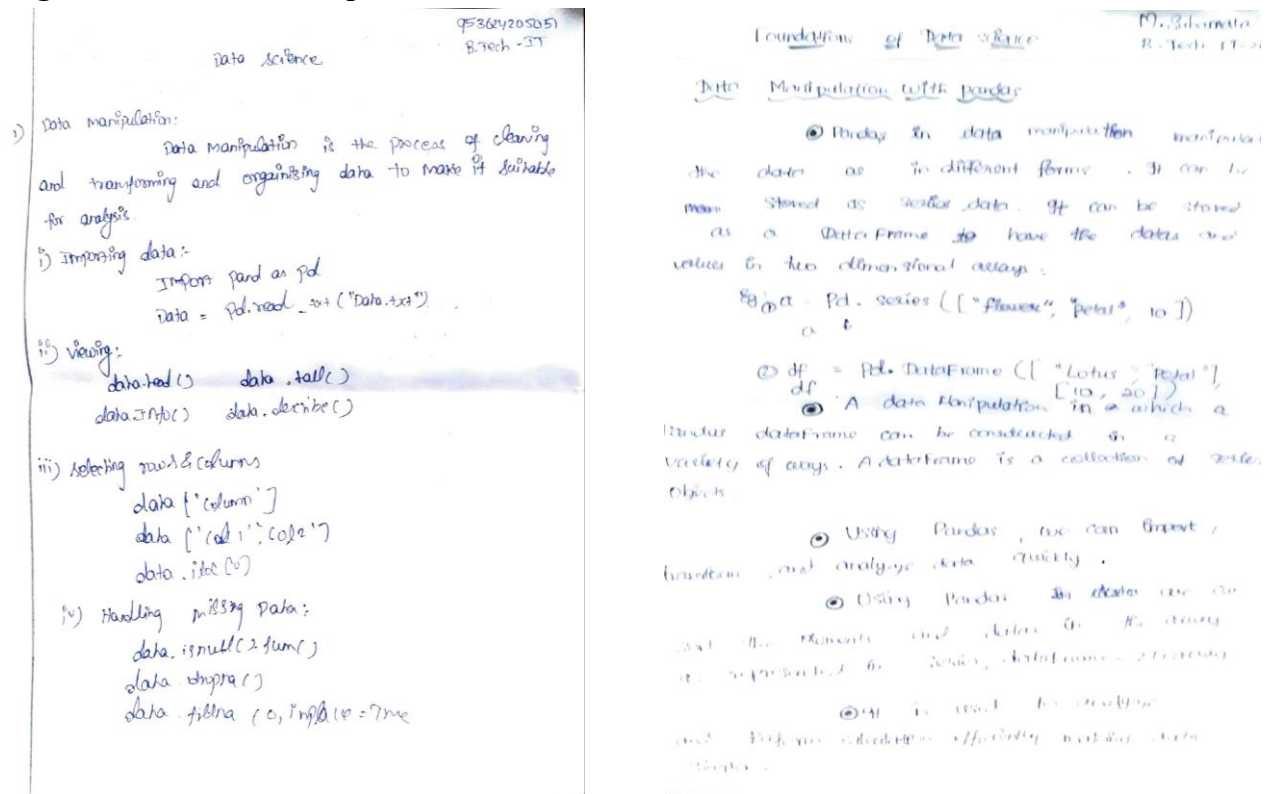


Figure 1: One Minute paper

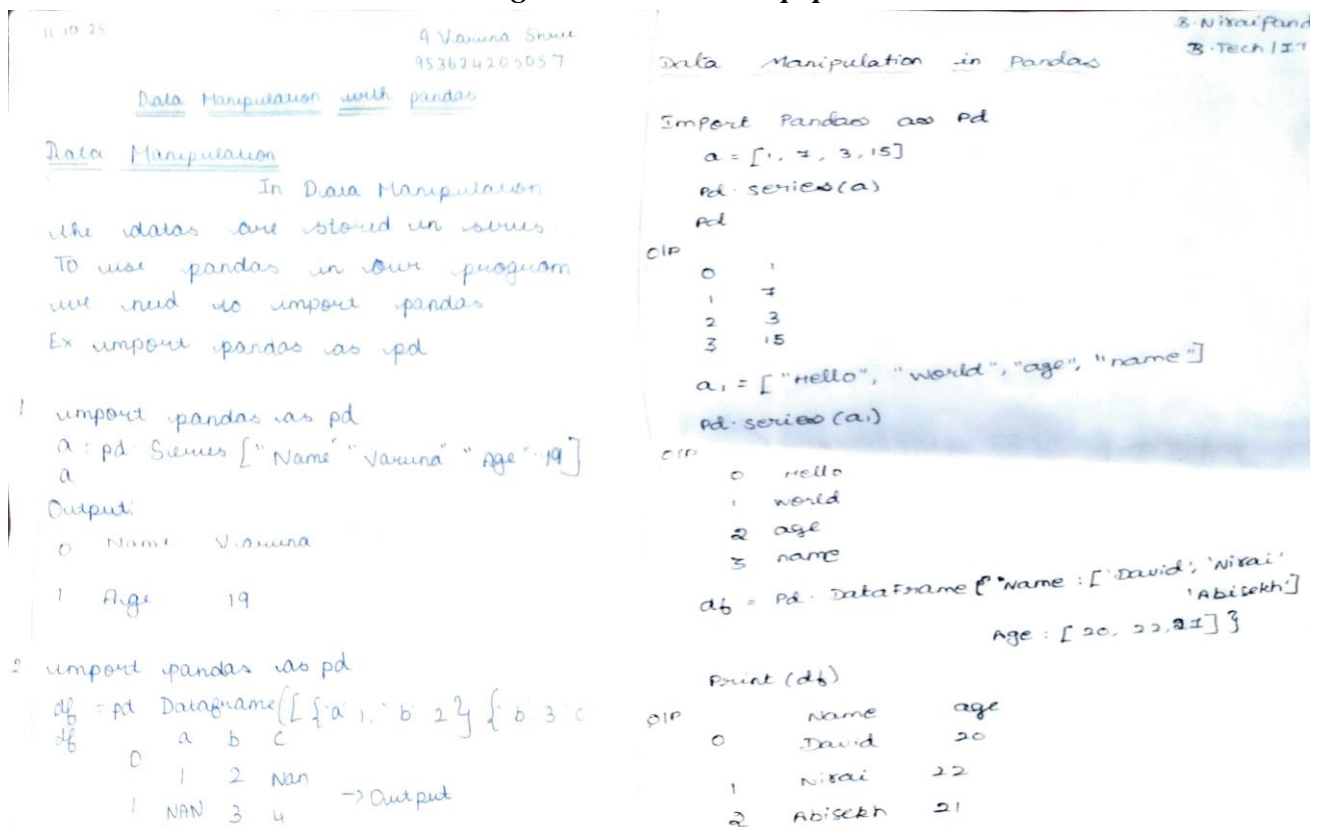


Figure 2 One Minute Paper Activity

Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***
 - Students found the one-minute practice activity engaging and helpful in reinforcing key Pandas operations such as data filtering, grouping, and handling missing values
 - A few suggested including slightly more complex datasets to better simulate real-world data scenarios.
 - Stakeholders suggested incorporating brief peer discussion or reflection at the end to enhance collaborative learning and conceptual clarity.
- ***Benefit of the practice:***
 - The one-minute paper activity enabled students to quickly reflect on their understanding of key Pandas concepts such as data cleaning, filtering, grouping, and aggregation.
 - It encouraged active participation, critical thinking, and self-assessment among students.
- ***Challenges faced in implementation:***
 - Some students initially found it difficult to summarize their understanding within a short time frame, especially when dealing with multiple Pandas functions and syntax.
 - Limited time also restricted detailed reflection and feedback. A few students faced difficulty in expressing conceptual doubts clearly in writing.

References:

- https://www.ritrjpm.ac.in/images/B.Tech-IT/2024-2025/VA_CS3452_OneMinutePaper.pdf
- <https://www.ijfcm.org/article-details/13601>

Signature of Faculty Member**HOD**